```
# Import Required Libraries
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)

# ------------------------------------------
# Load Dataset
# ------------------------------------------

df = pd.read_csv("loan_data.csv")

# ------------------------------------------
# Remove Loan_ID Column
# ------------------------------------------

if "Loan_ID" in df.columns:
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```
    df.drop("Loan_ID", axis=1, inplace=True)

# -----------------------------------------
# Handle Missing Numerical Values
# -----------------------------------------

num_cols = [
    "ApplicantIncome",
    "CoapplicantIncome",
    "LoanAmount",
    "Loan_Amount_Term",
    "Credit_History"
]

for col in num_cols:
    df[col] = df[col].fillna(df[col].mean())

# -----------------------------------------
# Handle Missing Categorical Values
# -----------------------------------------

cat_cols = [
    "Gender",
    "Married",
    "Dependents",
    "Education",
```

```
    "Self_Employed",
    "Property_Area",
    "Loan_Status"
]

for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

# ------------------------------------------
# Encode Categorical Columns
# ------------------------------------------

encoder = LabelEncoder()

for col in cat_cols:
    df[col] = encoder.fit_transform(df[col].astype(str))

# ------------------------------------------
# Separate Features and Target
# ------------------------------------------

X = df.drop("Loan_Status", axis=1)
y = df["Loan_Status"]

# ------------------------------------------
# Split Dataset
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# -----------------------------------------

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# -----------------------------------------
# Random Forest Function
# -----------------------------------------

def randomForest(X_train, X_test, y_train, y_test):

    model = RandomForestClassifier(
        n_estimators=100,
        random_state=42
    )

    # Train Model
    model.fit(X_train, y_train)

    # Prediction
    y_pred = model.predict(X_test)
```

```
# Accuracy
train_accuracy = accuracy_score(
    y_train,
    model.predict(X_train)
)

test_accuracy = accuracy_score(
    y_test,
    y_pred
)

# Display Results
print("=" * 50)
print("Random Forest Model Results")
print("=" * 50)

print("\nTraining Accuracy :", round(train_accuracy * 100, 2), "%")
print("Testing Accuracy  :", round(test_accuracy * 100, 2), "%")

print("\nConfusion Matrix")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report")
print(classification_report(y_test, y_pred))
```

```
# -----------------------------------------
# Call Function
# -----------------------------------------

randomForest(X_train, X_test, y_train, y_test)
```